

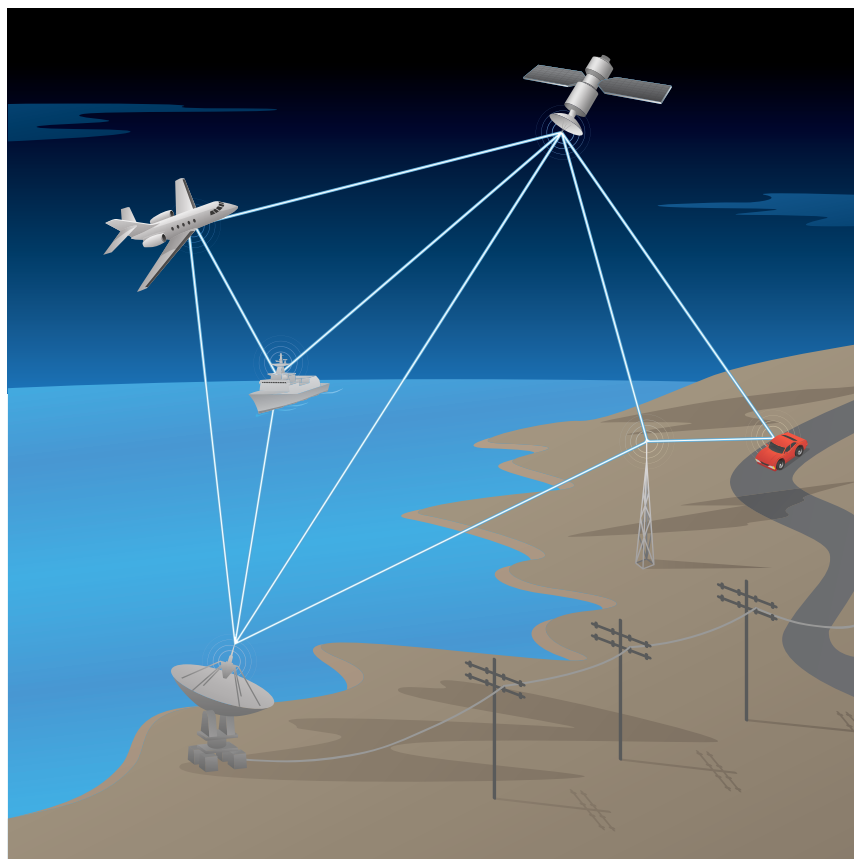
known as large constellations, adorns the celestial stage. These constellations consist of hundreds and thousands of satellites working in unison, gracefully orbiting the Earth as a synchronised system. Together, they aim to revolutionise global internet connectivity, offering low-latency and high-capacity access to users around the world.

While the advent of large constellations promises groundbreaking advancements in global communication, it also raises crucial legal concerns. One of the foremost issues is ensuring the cyber safety of these intricate systems. Safeguarding large constellations and their assets from potential cyber-attacks becomes paramount in this new era of interconnectedness.

As these constellations proliferate, it becomes essential to develop a robust legal framework that promotes the sustainable and secure use of outer space, including the large constellations, their associated technologies, and the services they provide. Achieving this delicate balance between legal regulations and technological ingenuity presents a fascinating challenge.

One significant area of concern is space safety. The deployment of satellites by various entities contributes to congestion in Low Earth Orbit (LEO), which, in turn, increases the risks of collisions between satellites in orbit. Such collisions can have far-reaching consequences and undermine the core principle of customary international law in outer space—the “freedom of access.” The freedom to operate and navigate space without undue restrictions is a fundamental tenet of space exploration, and preserving it amidst the surge of large constellations becomes a pivotal task.

Moreover, the vulnerability of satellites to cyber-attacks in space adds an additional layer of com-



*Satellite GPS network communication scene with aircraft, ship, ground antenna, and car*

plexity. While space assets were traditionally considered secure from physical aggression, the evolving landscape of cyber warfare now extends into the outer reaches of the cosmos. Protecting satellites from cyber threats and ensuring the integrity of their operations becomes an urgent matter that necessitates comprehensive legal provisions.

Interestingly, the existing space treaties, such as the Outer Space Treaty (OST), do not explicitly define the concept of a “large constellation of satellites.” This lack of precise terminology poses a unique challenge for regulators and policymakers. As these innovative systems transcend the boundaries of conventional space activities, existing legal frameworks must adapt to address the novel issues raised by large constellations.

Harmonising the governance of large constellations at the international level becomes a subject of captivating discussion. As different countries have varying space-related procedures and regulations, the challenge lies in establishing coherent and unified regulations that transcend national boundaries. Avoiding legal fragmentation while effectively governing large constellations becomes a crucial objective in this endeavour.

To ensure a secure and interference-free environment, preventing harmful interference between operators, is of utmost importance. The intentional disruption or blockage of satellite signals can have severe consequences, affecting vital services such as communication, navigation, and even endangering international air traffic. The Inter-